Cover letter

[PENDING: submission date]

To the Editor-in-Chief, *Science and Medicine in Football*

Dear Editor,

I submit the Original Investigation **"What can be measured from the broadcast? Validity of single-camera video for football-kick kinematics and swing-leg joint loads"** for consideration in *Science and Medicine in Football*.

Clubs, leagues and technology vendors increasingly derive technique and load metrics from broadcast video, and since the 2026/27 season LaLiga timestamps every touch with a connected ball. Whether the kicking body can be measured from the same broadcast has not been tested against a reference. This method-comparison study asks a practical question for football: for which uses — comparison with laboratory norms, ranking players or monitoring an individual — are kicking kinematics and swing-leg joint loads estimated from a single broadcast-like camera acceptable, and how large would a study validating a broadcast-derived load score as an injury predictor have to be?

The main findings are:

- From a single broadcast camera, none of six kinematic features or three swing-leg joint moments met literature-based acceptability criteria for any of the three uses. The detector error on 200 Premier League shots was 7.6–12.9 % of player height.
- Fusing the replay angles a broadcast already shows, with a laboratory-learnt correction, made kicking-knee flexion acceptable for ranking players and trunk leans comparable with laboratory norms. Knowing the contact instant, as the connected ball provides, reduced errors by a quarter; frame rate did not matter.
- A swing-leg Kick Load Index tracked kick intensity in the laboratory but was not recoverable from broadcast-like video. Validating it as an injury predictor would therefore need about 590 kicking-related injuries, 12–24 seasons of the whole league, against 48 with a laboratory-grade measurement — a quantitative argument for raising measurement reliability before any prospective injury study.

The methods were chosen to support these conclusions without overstating them. Ground truth comes from optical motion capture of 90 kicks, and the error model is the real detector error measured on broadcast shots, not idealised noise. The simulation is tested against real video with synchronised capture and a per-recording digital twin ($r = 0.84$). Agreement is reported following GRRAS, with equivalence tests and statistical parametric mapping. A field example with the author's own ball kicks filmed by smartphone shows the same patterns on real football kicks. Limitations are stated explicitly, in particular that most reference kicks are laboratory air kicks. All code and the broadcast residual bank are openly available (https://github.com/convderover/broadcast-kick-biomechanics; Zenodo DOI 10.5281/zenodo.23439408), and the complete statistical outputs and the GRRAS checklist are provided as supplementary files.

In line with Taylor & Francis policy, a preprint of this manuscript has been posted at **[PENDING: preprint server and DOI]**. The study is a secondary analysis of public datasets and was not pre-

registered. The extensive use of generative AI (study design proposals, code, analyses, figures and drafting) is declared in detail in the manuscript, and all code and outputs are public.

The manuscript is original, has not been published and is not under consideration elsewhere. I am its sole author, I report no competing interests and the work received no funding.

Thank you for considering this work.

Yours sincerely,

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